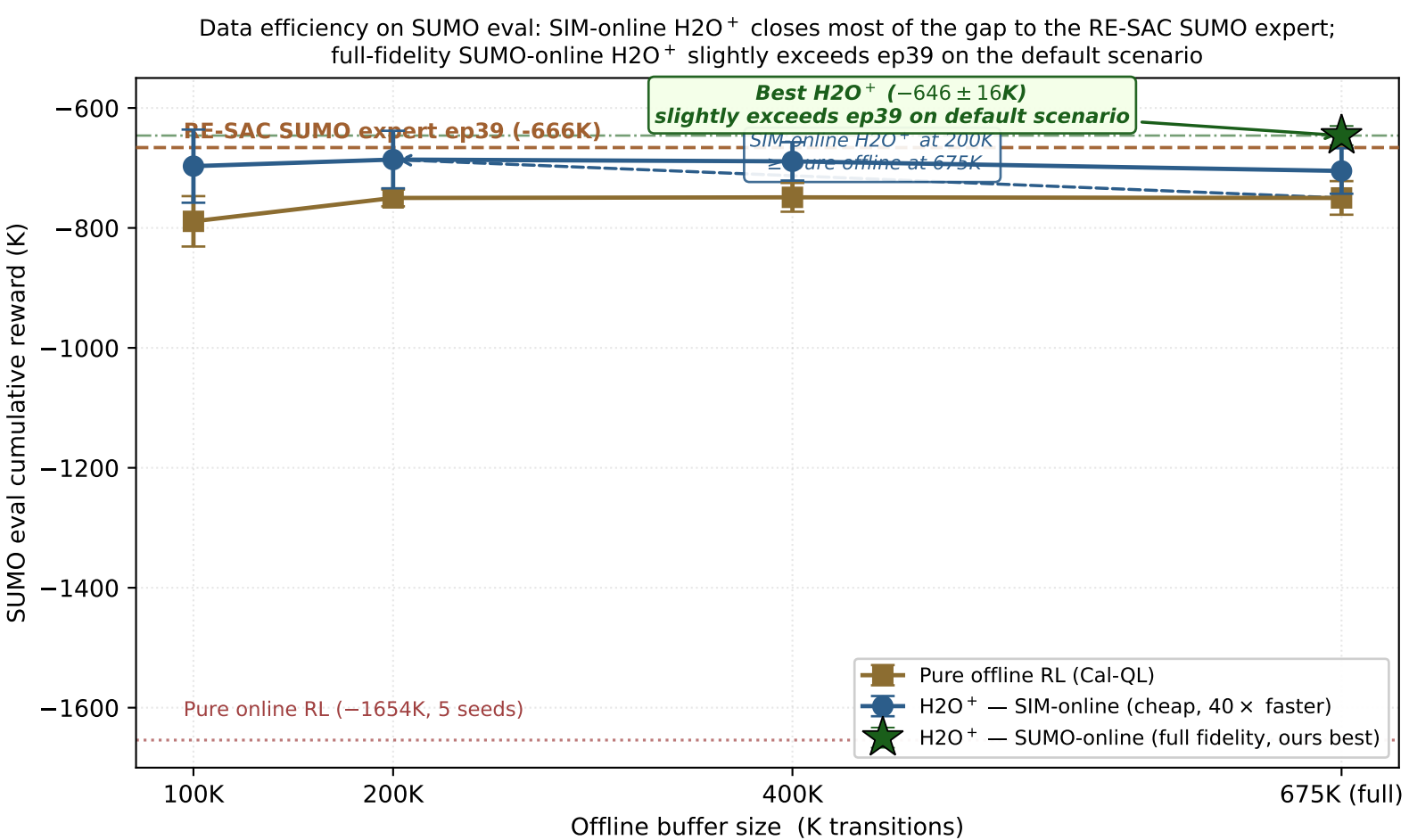
Data efficiency on SUMO eval: SIM-online H2O+ closes most of the gap to the RE-SAC SUMO expert;
full-fidelity SUMO-online H2O+ slightly exceeds ep39 on the default scenario
Best H2O+ (−646 ± 16K)
slightly exceeds ep39 on default scenario
RE-SAC SUMO expert ep39 (-666K)
SIM-online H2O+ at 200K
≥ pure offline at 675K
Pure online RL (−1654K, 5 seeds)
Pure offline RL (Cal-QL)
H2O+ — SIM-online (cheap, 40× faster)
H2O+ — SUMO-online (full fidelity, ours best)
SUMO eval cumulative reward (K)
−600
−800
−1000
−1200
−1400
−1600
100K
200K
400K
675K (full)
Offline buffer size (K transitions)